

Comparative Analysis of PCA, UMAP, and VAE for 3D Ear Mesh Dimensionality Reduction

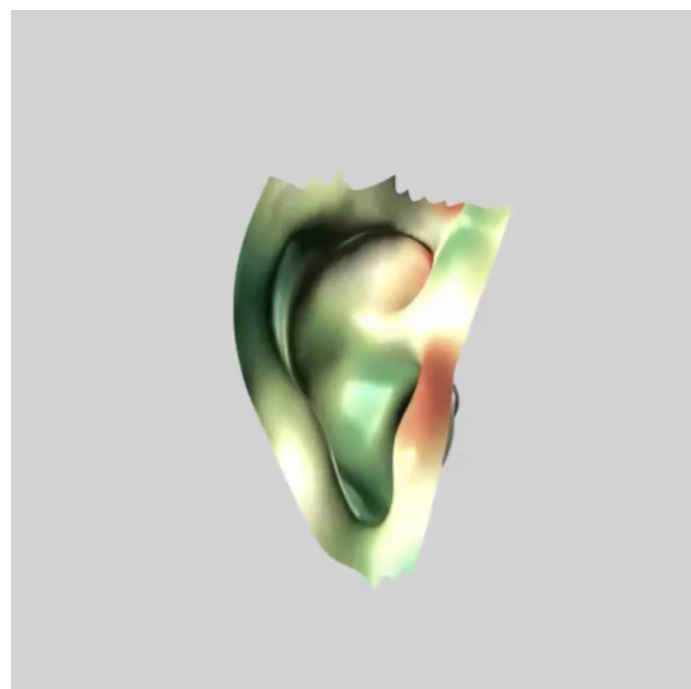
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Project Overview

Designing audio devices that fit a wide range of human ear shapes is a major challenge: comfort and performance vary with ear geometry. I analyzed a large dataset of 3D ear scans to determine what dimensionality reduction method—PCA, UMAP, or VAE—best captures population-level variation for reconstructing realistic average ear shapes. Given that PCA, a linear model, may struggle to capture the complex, intrinsic 3D geometry of the ear, I included the non-linear methods UMAP and VAE for comparative evaluation. Accuracy was evaluated using key anatomical features, including cavum width, height, depth, and tragus–antitragus distance (mm).



Rendering of Ear Data.

Data Pipeline

This pipeline processes large-scale 3D ear mesh data:

- Data Ingestion**
Pulled 3D ear mesh code from GitHub and loaded onto HPC cluster.
- Sampling**
Generated and stored 10,000 unique ear meshes, resulting in a large NumPy array of size (9760, 3, 10,000).
- Feature Extraction**
Computed and stored key anatomical measurements (4 features: cavum width/height/depth, tragus–antitragus distance) as a (10,000, 4) NumPy array for downstream evaluation.
- Dimensionality Reduction**
Applied PCA, UMAP, and VAE to reduce dimensionality across latent space ($n = 1, \dots, 30$).
- Error Computation**
Used Mean Squared Error (MSE) to compare reconstructed anatomical values against original measurements for each latent dimension.
- Visualization**
Generated graphs summarizing reconstruction accuracy and trends using Matplotlib.

All scripts and workflows were version-controlled via Git to ensure reproducibility and track changes.

PCA (Principal Component Analysis)

PCA with Singular Value Decomposition (SVD) was applied to reduce dimensionality while preserving key geometric features.

- The 3D ear meshes were flattened into a large 2D array of shape (10,000 × 29,280) to fit the PCA mod
- Data was prepared using Z-score normalization (centering and scaling), and PCA was then applied across latent dimensions $n = 1, \dots, 30$.
- Reconstruction was performed using the inverse PCA transform and reversed normalization to return data to its original dimensions for accuracy evaluation.

UMAP (Uniform Manifold Approximation and Projection)

UMAP was used to reduce high-dimensional 3D ear meshes, offering a nonlinear approach to capture more complex local and global geometric structure than PCA.

- The 3D ear meshes were flattened into a 2D array of shape (10,000 × 29,280) to fit the model.
- UMAP was fitted for latent dimensions $n = 1, \dots, 30$. The low-dimensional embedding was optimized using Stochastic Gradient Descent (SGD) to preserve neighborhood relationships.
- Reconstruction to the original dimensions was achieved using a supervised regressor model.

Note: Full-scale reconstruction over all latent dimensions required multiple days of HPC processing; the full workflow and intended evaluation are presented here.

VAE (Variational Autoencoder)

A Variational Autoencoder (VAE) was implemented to encode the high-dimensional data into a probabilistic latent space, enabling generative sampling and preserving geometric structure.

- The model was trained on the 3D ear meshes dataset (size (9760, 3, 10,000)) using a 70/10/30 training/validation/testing split. Data was batched by vertex position to the size of 50.
- The encoder consisted of three 1D convolutional layers (channels: $3 \rightarrow 64 \rightarrow 128 \rightarrow 1024$) that processed each vertex's (x, y, z) coordinates independently. Max pooling was applied after the final convolutional layer.
- The reparameterization trick was employed to sample the latent space (mean and variance) and enable backpropagation, while a mirrored Multi-Layer Perceptron (MLP) served as the decoder for reconstruction.

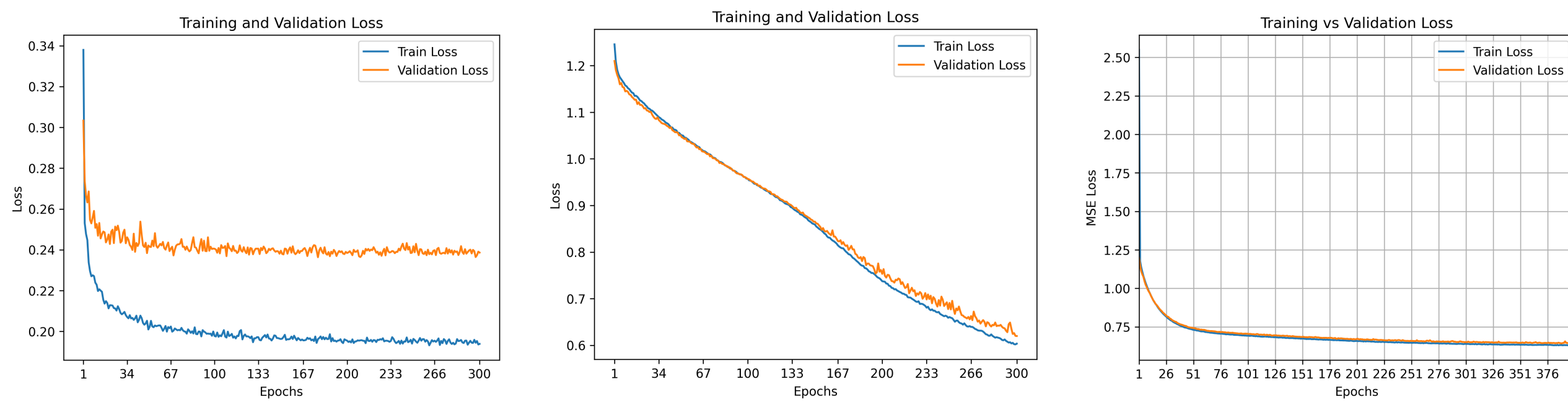


Figure 1. VAE training curves for Adam (10^{-3}), SGD (10^{-3}), and SGD (10^{-2}).

Training minimized the combined Mean Squared Error (MSE) and Kullback-Leibler (KL) divergence losses. After testing several configurations (including Adam and SGD), the final convergence was achieved using SGD with a learning rate of 10^{-2} , a batch size of 50, and 400 epochs.

As shown in the training curves, the SGD(10^{-2}) configuration achieved the steepest early decrease and the lowest final losses, with training and validation losses converging from 2.55 to 0.63 and 1.20 to 0.64, respectively. This demonstrated superior stability and convergence compared to Adam (10^{-3}) and other SGD configurations.

Error Analysis

PCA produced the lowest reconstruction errors at 21 principal components. Errors (MSE) for key anatomical measurements are shown below:

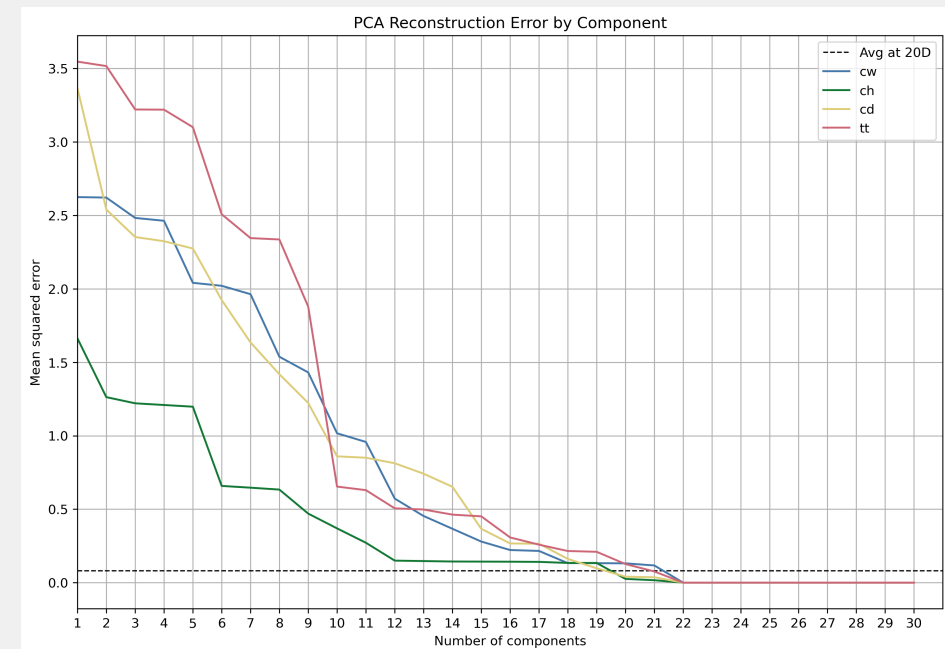


Figure 2. PCA MSE of anatomical measurements for each dimension

Feature	MSE (mm)
Cavum Width	0.1169
Cavum Height	0.0163
Cavum Depth	0.0366
Tragus–Antitragus Distance	0.0759

Table 1. PCA reconstruction errors at 21 principal components.

Error decreases as the number of PCs increases, converging around PC 22. Cavum height is consistently low in error; tragus–antitragus distance shows higher error due to variability.

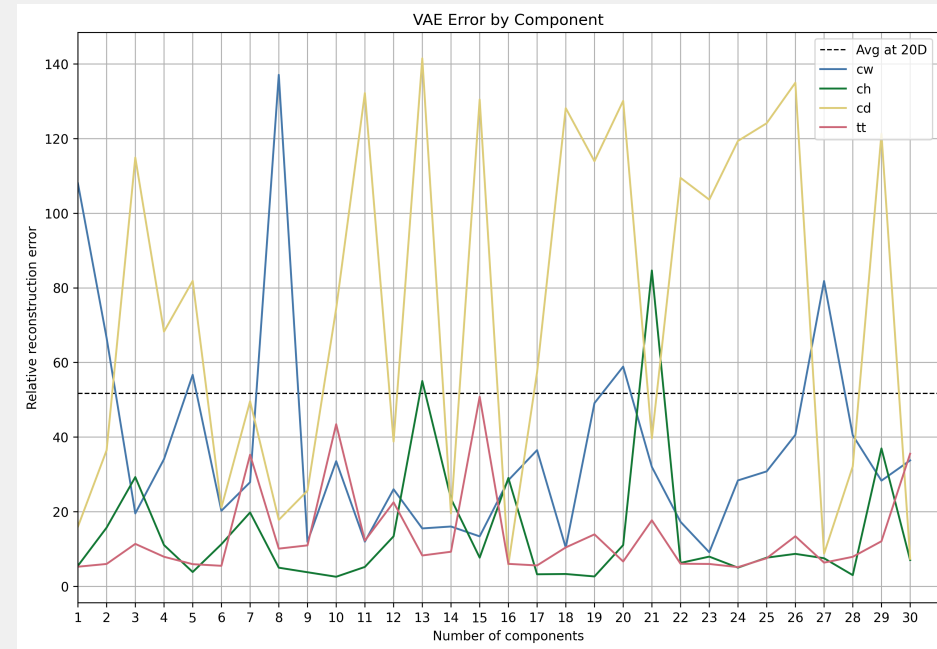


Figure 3. VAE MSE of anatomical measurements for each dimension

VAE results showed reconstructed error of anatomical measurements was high and oscillated across latent dimensions. This result was likely due inputs being batched by vertex position rather than by ear, causing the model to learn population-level variation of each vertex instead of coherent ear geometry. Then Max pooling amplified outlier vertices.

UMAP results could not be generated within the project timeline.

Conclusions

Key takeaways:

- PCA was the most effective and reliable dimensionality reduction method for 3D ear shape data.
- VAE and UMAP did not produce usable reconstructions under the configurations tested.
- VAE performance was limited primarily by incorrect batching strategy and use of max pooling, which disrupted geometric consistency.

Future improvements include: ‘

- Batch full ears rather than vertices in the VAE.
- Use average pooling in the VAE to preserve global geometry.
- Applying parametric UMAP for learnable embeddings.

References

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